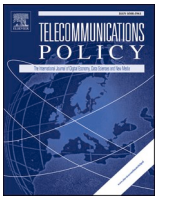
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Governing AI virtual anchors in China's live streaming E-commerce ecosystem: Policy challenges and global implications

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ABSTRACT

The rapid advancement of generative artificial intelligence (AI) has fundamentally reshaped the traditional media value chain, transforming the processes of content production, distribution, and consumption. Among these developments, AI virtual anchors have significantly reduced operational costs and enabled the large-scale creation of content. However, their widespread adoption has also raised complex legal, ethical, and regulatory challenges. This paper investigates the governance of AI virtual anchors from three key dimensions.

First, it examines how AI technologies are restructuring the media ecosystem, particularly in the realm of live-streaming e-commerce, by displacing human labour and creating new market dynamics. Second, it examines the associated legal and ethical concerns, including intellectual property disputes, the under-recognized rights of "ghost performers", risks of misinformation, and consumer protection issues. Third, it evaluates China's evolving governance responses, highlighting both proactive regulatory innovations and ongoing challenges. Starting from platform governance theories, this paper develops a China-specific regulatory narrative and identifies a multi-tiered governance system that involves the government, platforms, and public participation, and reveals the underlying logic that redefines platform roles in China's digital governance architecture. This paper argues that China's evolving governance of AI virtual anchors illustrates a distinct institutional model and aims to situate this experience within global discussions, offering comparative reference points for AI governance, particularly regarding platform responsibility, adaptive regulation, and public participation.

1. Introduction

The global virtual digital human industry is undergoing significant structural expansion, with market size projected to grow from USD 29.51 billion in 2022 to USD 527.58 billion by 2030 (Emerge Research, 2023).[2] Amid the broader transformation of the digital economy, the live commerce sector has exhibited significant technological innovation through the widespread adoption of virtual anchors, which are redefining commercial value chains. According to McKinsey & Company's 2024 industry analysis,[3] live commerce

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[1] Research areas in AI governance, Children privacy protection, law and ethic in advertising industry.

[2] Digital Human Avatar Market, https://www.emergenresearch.com/industry-report/digital-human-avatar-market.

[3] McKinsey& Company, How live commerce is transforming the shopping experience, https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/its-showtime-how-live-commerce-is-transforming-the-shopping-experience.

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is anticipated to contribute 10 %–20 % of global e-commerce sales by 2026, marking a new phase in media monetization models.

China occupies a leading position in this domain, offering a representative case study that illustrates the complex interplay between AI-driven innovation, platform governance, and regulatory adaptation. With a user base of 833 million (CNNIC, 2024) and transaction volumes increasing from 0.196 trillion yuan in 2017 to 4.9 trillion yuan in 2023 (Wangjingshe, 2023),[4] China's live streaming e-commerce ecosystem exemplifies rapid digital transformation. The deployment of AI virtual anchors, such as the AI avatar of Jingdong CEO Liu Qiangdong, which generated sales exceeding 50 million yuan during the 2024 "618 Shopping Festival," demonstrates the convergence of advanced technological capabilities and heightened consumer engagement.[5] As of October 2023, China had nearly 140 million live streaming accounts engaged in various activities, including video performances, with virtual live streamers constituting approximately 40 % of this total, surpassing 50 million in number (Global Times, 2023).

Despite the rapid expansion of AI-driven live streaming e-commerce, this growth has exposed systemic tensions within the AI media ecosystem. While AI-powered anchors enhance continuous engagement and offer personalized user experiences, they simultaneously disrupt traditional value chains, raising significant concerns regarding labour displacement, intellectual property rights (IPR), algorithmic transparency, and ethical implications (Deepak, 2024; Noor M. Alkudah & Tareq O. Almomani, 2024; Chi, 2024). Notably, the increasingly indistinct boundary between human and AI virtual anchors complicates issues of accountability, particularly in cases of false advertising. The regulatory data from the State Administration for Market Regulation shows that complaints and reports related to live streaming sales on the national 12315 platform increased by 52.5 % year-on-year in 2023, representing a 47.1 % rise over the past five years.[6]

The ethical and regulatory challenges emerging from AI-driven live streaming e-commerce are not isolated phenomena; they are a microcosm of the broader global concerns surrounding AI governance (Tuzov & Lin, 2024). Issues such as the ambiguity of legal liability, the non-transparent collection of consumer data during interactions, and the potential for algorithmic manipulation of purchasing behavior reflect worldwide debates on accountability, privacy (Ye et al., 2024), and digital consumer rights (Crawford, 2021). Therefore, studying AI virtual anchors provides a tangible and concentrated case study, offering critical insights into complexities of the digital economy (Flew, 2021), thus highlighting the profound importance of this research area.

In response, China adopts a hybrid framework that emphasizes national security, public order, and platform responsibility. Drawing on Gillespie's (2018) platform governance theory, this study argues that China has developed a unique, multi-tiered regulatory structure that integrates government regulation, platform responsibility, and public participation. However, this model still has unresolved issues. Accordingly, this paper investigates the transformative role of AI virtual anchors in reshaping China's live streaming ecosystem and explores its implications for global AI governance. Beyond the broader transformation of the digital economy, this paper also engages directly with the theme of AI and digital media ecosystems. The Chinese case of AI virtual anchors connects national governance practices to wider theoretical discussions on platform governance and media ecology.

Utilizing case studies and thematic policy and document analysis, the research examines.

1. The impact of AI anchors on value chain dynamics, including content production, distribution and consumption;
2. The legal and ethical challenges posed by AI virtual anchors within the live streaming media ecosystem, particularly concerning intellectual property rights (IPR), personality right infringement, false advertising, and broader ethical and social responsibility concerns;
3. Chinese multi-tiered AI virtual anchor governance strategies (from government, platform and public participation) and policy dilemmas.

This paper is structured into nine sections. Section 2 presents the literature review and theoretical framework, combining platform governance theory with the Chinese regulatory context to establish an analytical foundation. Section 3 outlines the research methodology, which integrates case study analysis and thematic policy and document analysis. Section 4 addresses Research Question One by tracing the development trajectory of AI virtual anchors in China and examining how their emergence has restructured the traditional media value chain. Section 5 addresses Research Question Two by examining the legal and ethical risks associated with AI virtual anchors, including copyright infringement, personality rights violations, false advertising, and ethical concerns. Sections 6 and 7 jointly respond to Research Question Three by providing a detailed account of China's multi-tiered governance framework, which comprises government regulation, platform responsibility, and public participation, and identifying key institutional and operational challenges within it. Section 8 offers a comparative perspective by examining China's governance of AI virtual anchors alongside platform governance theories which developed in a more industry-led regulatory context. It analyses how regulatory responsibilities are distributed among the government, the platform, and the public under a context-specific, multi-tiered governance framework. Section 9 concludes the paper by summarizing the key findings, acknowledging the study's limitations, and suggesting directions for future research. The structure is shown in Fig. 1.

[4] Wangjingshe (2023). 2023年度中国直播电商市场数据报告 [2023 China live e-commerce market data report]. https://www.100ec.cn/detail–6639773.html.

[5] China's JD.com ramps up AI use on its platform as founder Richard Liu Qiangdong debuts avatar in live streaming. sessions, https://www.scmp.com/tech/big-tech/article/3259232/chinas-jdcom-ramps-ai-use-its-platform-founder-richard-liu-qiangdong-debuts-avatar-live-streaming.

[6] See China State Administration for Market Regulation official page, https://www.samr.gov.cn/xw/zj/art/2024/art_992bb54bd6694032bb9b60d91f4924e0.html.

2. Literature review and theoretical framework

As AI technologies increasingly shape digital communication, the rise of virtual anchors in China's live streaming e-commerce sector presents complex governance challenges involving legal status, ownership, cultural expression, and ethical responsibility. This literature review focuses on four key areas: content ownership, cultural rights, ethical dilemmas, transparency and accountability, and the risks of homogenization and technological dependency.

2.1. Literature review

Ownership and Authorship of AI-Generated Content (AIGC). AI systems challenge the traditional concept of authorship. Boden (1998) conceptualized "computational creativity," suggesting that AI can achieve novelty through combinational, exploratory, and transformational mechanisms. While most jurisdictions still attribute legal authorship to humans (Fritz, 2025; Xiao, 2022), the ownership of AI-generated products remains a disputed issue. McCormack et al. (2019) argued that copyright may need to be shared among artists, algorithm developers, and data contributors. In the case of AI virtual anchors, Xu and Li (2024) proposed that potential rights holders could include software developers, platform operators, or commissioning entities. However, the absence of clear legal definitions continues to hinder regulatory enforcement and complicates the attribution of liability.

Cultural Ownership in Contextualized AI Expression. Beyond legal authorship, cultural ownership emphasizes collective and symbolic rights to language, aesthetics, and behaviour (Ba'ai and Aris, 2024). Robinson (2020) observed that national and institutional cultures play a pivotal role in shaping the governance of artificial intelligence. In Western discourse, cultural appropriation is commonly discussed in the context of fashion, entertainment, and tourism, where dominant groups commodify marginalized cultural

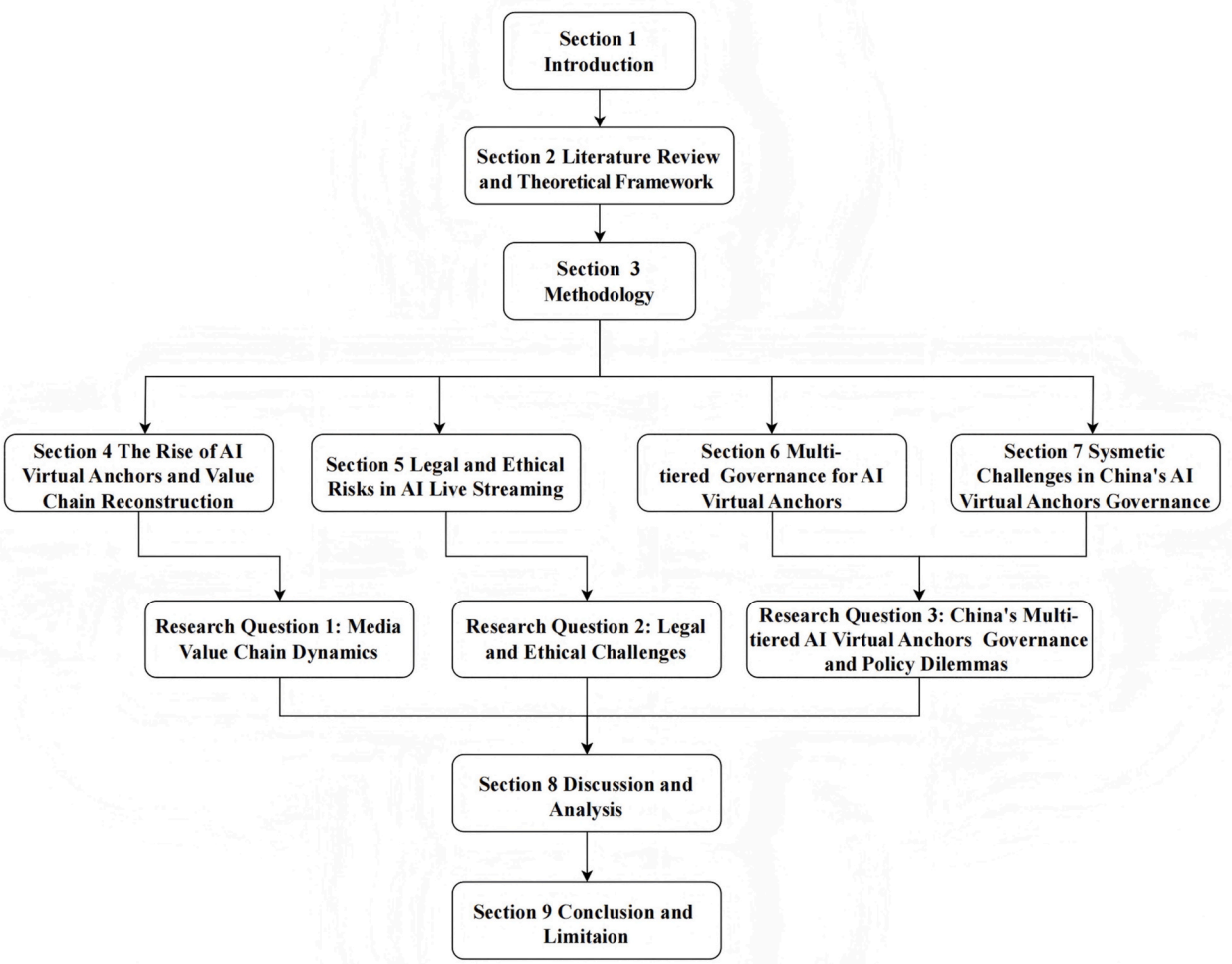


Fig. 1. Structure of the paper.

elements (Kawamura & de Jong, 2022; Rogers, 2006). In China, such issues are more prominently manifested as internal cultural misrepresentation, involving elements such as synthesized dialects, Hanfu-inspired aesthetics, and regional idioms. While these features contribute to cultural localization and user engagement, especially with the integration of artificial intelligence into cultural dissemination, they have also raised concerns regarding the misuse (Cao, 2025) and stereotyping of minority cultures (Ge et al., 2024), highlighting the need for content governance.

Ethical Considerations in AI Live Streaming. Ethical issues surrounding AI have long been debated. Asimov's "Three Laws of Robotics" provided an early conceptual foundation for ethical considerations in human-machine interaction (Asimov, 1942). Contemporary scholarship, notably by Floridi and Cowls (2019), has since expanded the discourse to encompass issues of trust, authenticity, accountability, algorithmic bias, and broader societal impact. Within the context of live streaming, AI virtual anchors may be implicated in ethically problematic practices, including false advertising, face-swapping fraud, and the dissemination of inappropriate content to minors. Transparency and explainability (Balasubramaniam et al., 2023), crucial for algorithmic governance (Cao, 2024; Shukla, 2024), remain hampered by the "black box" nature of algorithms (Pasquale, 2015). Inconsistent content labeling practices across platforms further obscure responsibility, weakening public trust and ethical safeguards.

Risks of Content Homogenization and Technological Dependence. Algorithm-driven content production is reshaping live streaming norms, but it is also exacerbating creative stagnation and systemic risk. Pariser's "filter bubble" theory highlights how personalized recommendation systems can promote informational isolation (Pariser, 2011). In the Chinese context, many AI virtual anchors adopt standardized visual templates and scripted marketing narratives, resulting in significant content homogenization (Ghaith, 2024), such as the frequent repetition of slogans like "lowest price on the internet" (Liu et al., 2025). Moreover, excessive reliance on AI technologies risks displacing human anchors. Some companies employ human actors behind virtual avatars, so-called "ghost performers", whose contributions remain largely invisible and unacknowledged. This raises additional concerns regarding equity and labour recognition.

While existing scholarship has established foundational theoretical frameworks for artificial intelligence, substantial gaps persist when these are applied to culturally embedded, domain-specific practices such as AI-powered live streaming. Situated within China's technological and policy context, this study investigates AI virtual anchors within the live streaming e-commerce ecosystem, with particular attention to the impact of AI virtual anchors on value chain dynamics, legal and ethical issues and China's governance dilemmas.

2.2. Theoretical framework

To analyze the governance structure of AI virtual anchors in China's live streaming ecosystem, this study adopts a comparative framework that integrates platform governance theory with contextualized regulatory practices. It draws on Gillespie's (2018) notion of platforms as "reluctant regulators" and van Dijck et al.'s (2018) portrayal of platforms as infrastructural institutions, perspectives highlighting platforms' dual functions in shaping public discourse while resisting direct government control.

While Gillespie's theory informs studies on platform governance in China (e.g., Yan & Du, 2022, 2024; Lu et al., 2022), its application within the country's government-guided framework remains underexplored. This study extends Gillespie's model by situating it within a regulatory environment where platforms are embedded within government-guided regulatory frameworks. In doing so, the study adapts the original theory to a distinct governance context—one in which platforms are more formally integrated into national regulatory objectives and assume roles aligned with broader policy coordination.

More specifically, this study develops a conceptual framework for understanding the platforms' dual roles in China's live streaming ecosystem. It demonstrates that platforms function both as policy-implementing regulators—responsible for executing technical requirements like real-name registration, algorithm filings, and AI content labeling—and as entities that are themselves regulated, obliged to comply with regulatory standards, engage in compliance reporting, and facilitate public participation. This dual positioning is dynamic, shaped through evolving administrative procedures, regulation arrangements, and formalized accountability structures that characterize the governance approach in China's digital domain (Zhang, 2021).

Furthermore, China's approach to AI governance emphasizes a dialectical balance between principle and flexibility (Shen, 2024), combining legal rigidity (like Cybersecurity Law and Data Security Law) with adaptive regulatory instruments such as pilot initiatives, and multi-actor participation (including government officers, platform leaders, scholars, online users, and other groups for discussing problems together).

By situating platform governance within China's institutional environment, this study retheorises Gillespie's theory, offering a contextualized contribution to global AI governance discourse.

Table 1
Overview of primary cases.

Case Number	Case Content	Legal/Ethical issue
Case 1	Shanghai, Shinchoho Co., Ltd. vs Hangzhou Jellyfish Intelligence Technology Co., Ltd.	Copyright infringement
Case 2	Mofa (Shanghai) Information Technology Co., Ltd. and Hangzhou Sihai Optical Fiber Network Co., Ltd.	Ghost performer labour dispute
Case 3	AI deepfake of Doc. Zhang Wenhong	Infringement of celebrity portrait rights and false promotion
Case 4	Douyin account"Russia Beauty"impersonates a Ukrainian YouTuber	Identity theft for false promotion
Case 5	Bilibili virtual anchor fabricated story for public attention	Malicious marketing
Case 6	AI virtual anchors scamming senior citizens	Mostly ethical problems
Case 7	AI virtual anchors spreading violent or obscene content to minors	

3. Methodology

To fully explore the core research questions of value chain restructuring, legal and ethical challenges, and governance mechanisms of AI virtual anchors within China's live streaming e-commerce ecosystem, a qualitative research design is adopted that combines case study analysis with thematic policy and document analysis.

At the same time, these approaches have certain limitations. They rely primarily on policy and regulatory documents and case studies, without incorporating interviews or quantitative data. While such document-based and case-oriented strategies provide a systematic and coherent understanding of the governance framework, they inevitably narrow the empirical scope of analysis. Future research could build on this foundation by incorporating fieldwork, stakeholder interviews, or large-scale data analysis to further enrich and substantiate the findings.

3.1. Case study analysis

This method enables a focused examination of judicial rulings involving AI virtual anchors. Case selection was guided by the following criteria: (1) direct involvement of AI-generated avatars in content-related legal or ethical disputes; and (2) representativeness in terms of regulatory response, public controversy, or the presence of evident regulatory gaps and unresolved legal issues (see Table 1).

3.2. Thematic policy and document analysis

Legal and policy documents. To systematically examine the regulatory architecture governing AI virtual anchors in China, this study employs a thematic content analysis approach. Legal and policy documents are categorized according to institutional level, including national legislation, administrative regulations, and implementation guidelines. These are analyzed across four key dimensions relevant to AI virtual anchors governance: regulatory focus, governance roles, main value orientation, and inter-institutional relationships. This analytical structure avoids a fragmented presentation of individual policies and instead emphasizes their systemic coherence and integrated logic (see Table 2).

By analyzing these interlinked regulatory layers, the study reveals an underlying coherence in China's governance logic: national laws establish broad normative principles, while administrative provisions address AI-specific risks and provide operational guidance for platform-level practices. Collectively, these documents form the foundational framework for the regulation of AI applications.

Platform Governance Practices. To complement China's unique regulatory framework, this study also examines the platform-level governance structures and enforcement mechanisms adopted by major Chinese live streaming platforms. While there is currently no official statistic quantifying the number of platforms using virtual anchors, it is evident that a growing number of platforms across news, commerce, and entertainment sectors are deploying AI-generated anchors.

This study focuses on five representative platforms—Douyin, Kuaishou, Bilibili, Douyu, and Taobao—selected for their scale, public influence, and regulatory engagement. Except for Douyu, which is cooperating with DeepSeek, the other platforms have developed and deployed proprietary AI models, as detailed in Table 3.

All the selected platforms operate within the same national regulatory framework, particularly the *Interim Measures on Generative AI (2023)* [7] and *the AI Content Labelling Measures (2025)* .[8] These regulations mandate avatar registration, algorithm filing, and human oversight of AIGC. However, enforcement practices vary across platforms. For example, Douyin and Taobao proactively prohibit fully AI-driven live streaming without human supervision, especially in consumer-facing scenarios. Whereas Kuaishou and Douyu implement more contextualized approaches, permitting specific AI-assisted formats in low-risk domains with real-time oversight. These platforms tend to adopt less strict approaches, emphasizing traffic moderation or context-specific content categorization.

[7] See China Interim Measures for Generative AI Services, https://www.gov.cn/zhengce/zhengceku/202307/content_6891752.htm.
[8] See Measures for Labeling of AI-Generated Synthetic Content, https://www.gov.cn/zhengce/zhengceku/202503/content_7014286.htm.

Table 2
Primary policy, laws and regulation related to AI virtual anchors.

Type	Policy/Law/Regulation/Document	Regulation Focus	Governance Role	Main Value Orientation	Inter-institutional relationships
National policy (National Strategic documents)	Principles for Governance of New Generation Artificial Intelligence: Developing Responsible AI (AI Governance Principles) (2019)	Top-level strategic guidelines for AI ethics, accountability, and governance	Government-guided	Public interest-oriented governance& controllability & sustainability, accountability & transparency, fairness & inclusiveness	Provide value orientation and ethical principles to guide subsequent legislation and governance goals
Laws (upper-level legislation)	Civil Code	Personality rights protection (portrait rights, voice rights)	Government-guided	Personal rights protection, prevention of identity theft	Establish basic institutional norms and core principles such as citizen rights, platform responsibilities, data sovereignty and so on
	Personal Information Protection Law (PIPL)	User data collection and processing	Government-guided	Privacy protection, principle of minimum necessity	
	Copyright Law	Authorship, ownership and Copyright Attribution	Government-guided	Intellectual property balance, benefit distribution	
	E-Commerce Law	Compliance of virtual anchors' live streaming Marketing Practices	Government-guided	Consumer rights and interests, market fairness	
	Cybersecurity Law	Cybersecurity of Online Content and Platform Accountability	Government-guided	National security, a clean cyberspace	
	Data Security Law	Data classification & hierarchical protection system	Government-guided	Data sovereignty, controllable risks	
Administrative policy (Administrative strategic documents)	The 14th Five-Year Plan for Science and Technology Development in Radio, Television, and Online Audio-Visual (14th Five-Year RTV-Audio-Visual S&T Plan)	Encouragement of AI virtual anchor development	National Radio and Television Administration (NARTA)	Technology innovation, Industry empowerment	Strategic Guidance & Industry Empowerment
Administrative Regulations (Implementation layer)	Interim Measures for the Management of Generative AI Services (Interim Measures on Generative AI)	Comprehensive AI governance (data training, algorithm, labelling, etc.)	Jointly issued by multiple departments: Cyberspace Administration of China (CAC); National Development and Reform Commission; Ministry of Education and other relevant departments	Inclusive and prudent AI regulation	Refine the implementation standards to respond to specific risks in generative AI scenarios
	Measures for Labeling of AI-Generated Synthetic Content (AI Content Labeling Measures)	Deep synthesis technology filing and traceability	Jointly issued by multiple departments: CAC; Ministry of Industry and Information Technology (MIIT); Ministry of Public Security (MPS)	Technologically controllable, traceable, accountability	
	Code of Conduct for Online live streaming Anchors (Online Anchor Conduct Code)	Human anchors' behaviour boundaries (such as speech, values)	NARTA	Ideological safety, core values	Specifies conduct standards for human live streamers; aligned with ideological and ethical norms
Industry Standard (technical practice)	Digital Virtual Human Technical Specifications	Classification, scenario-based application, avatar realism, driving technologies, platform capabilities, and safety thresholds	NARTA	Technical reliability and risk control, ethical compliance, and industry standardization	Clarify technical thresholds, application scenarios, virtual human classification standards and safety requirements, and technical details

Table 3
Overview of primary platform regulations.

Platform	Type	Monthly Active[a] Users (million)	AI models	Main Review Mechanisms	Regulation Highlights
Douyin	Short video	1000	Doubao, Huoshanfangzhou	Real-name and image registration, manual/algorithmic review	Prohibits unauthorized AI avatars, mandates registration and labeling, bans fully AI-driven live streams
Kuaishou	Short video	573	NVWA, Kling	Restricts traffic to low-quality AIGC live streams	Only Kuaishou official avatars allowed; no mechanical repetition
Douyu[b]	Game live streaming	44.5	Deepseek (Collaboration)	Real name and image registration, no copyright problems, no violence or obscenity	Virtual anchors must select"Virtual Entertainment" or "Anime"category
Bilibili	Long video	220	Bijian Studio	Strict content and design norms	Tight control on vulgar/malicious content; human-driven AI avatars only
Taobao[c]	Shopping	910	Tongyiqianwen	Private stream only for third-party AI anchors	Official AI avatars allowed in public streams only; pre-approval required

[a] QuestMobile, The monthly active users (MAU) of Douyin have exceeded 1 billion,https://news.china.com/socialgd/10000169/20250430/48280927.html According to QuestMobile, till 2025, the monthly active users(MAU) of douyin is 1000 million, kuaishou 573 million, bilibili 220 million.

[b] Sina Finance, 2025, DouYu Ramping Up Its Second Growth Engine,https://finance.sina.com.cn/jjxw/2025-05-21/doc-inexiptx5901699.shtml, according to douyu 2025 Q1 Financial Report, till 2025, the MAU of Taobao is 44.5 million.

[c] QuestMobile, Alibaba's Consumer Finance Expansion Gains Further Momentum,https://finance.sina.com.cn/money/bank/dsfzf/2025-05-26/doc-inexwtay1021734.shtml, as QuestMobile shows that till 2024, the MAU of Taobao is 910 million.

4. The rise of AI anchors and value chain restructuring

4.1. The rise of AI anchors

As McLuhan (1964) insightfully noted, "the medium is the message," and the evolution of media forms has repeatedly reshaped communication and social organization. The trajectory of virtual anchors—from Max Headroom to Hatsune Miku, Microsoft's Xiaobing, and today's AI-driven livestreaming anchors, illustrates this shift. As Meyrowitz (1985) argued, new media create new "social situations" and reconfigure roles between frontstage and backstage actors, presaging broader changes within the media ecosystem.

The development of AI virtual anchors has progressed through three main stages (Guo & Huang, 2022; CSDN, 2022). Prior to 2016, AI capabilities remained relatively underdeveloped. The period from 2016 to 2020 witnessed rapid advancements in AI technologies, while from 2020, the field has entered a new phase marked by increasing sophistication and integration. In the 1980s, the United Kingdom saw "the first computer-generated" TV program host, Max Headroom, marking the pre-interactive era of synthetic media. Around 2007, popular Japanese virtual idol Hatsune Miku emerged, primarily through animation and voice, generating annual revenues exceeding 10 billion Japanese yen (SankeiBiz, 2012). However, these early virtual idols lacked real-time interactivity and were largely dependent on pre-rendered animation and synthetic voice technologies.

From 2016, advances in AI and computing enabled the deployment of virtual humans across various domains. A notable example is Microsoft's Xiaobing, which, by 2018 had reached 5 million daily active users, exemplifying the potential of AI-driven virtual agents to engage and sustain interactive user experience.

Since 2020, the rise of the metaverse and virtual reality has accelerated the use of AI-driven virtual anchors in livestreaming and brand collaborations. A notable example is Ling Yanhuan, a virtual anchor inspired by the traditional Chinese nüxia (female martial hero), who has attracted 899,000 followers on Douyin and generated millions in revenue through commercial partnerships, as illustrated in Fig. 2. She exemplifies the integration of cultural identity and commercial utility in China's AI live streaming ecosystem (China Daily, 2023).

The rapid development of China's virtual anchor industry is closely linked to strong government support. As the foundation of China's hierarchical governance system, the government has played a dual role—driving early technological innovation while setting regulatory boundaries (Ma, 2024). The inclusion of AI virtual anchors in the National Radio and Television Administration's 14th *Five-Year RTV-Audio-Visual S&T Plan (2021)* [9] has facilitated their integration into program production (China Advertising Association Digital Metaverse Working Committee, 2021). This was followed by the 2024 release of *Digital Virtual Human Technical Specifications*,[10] which formally defined virtual humans and outlined regulatory standards by platform type and application scenarios. In response, giant Internet platforms such as Douyin and Taobao quickly launched proprietary AI systems aligned with national policy, further expanded the application scope and use scenarios of virtual anchors.

[9] See The 14th Five-Year Plan for Science and Technology Development in Radio, Television, and Online Audio-Visual, https://www.nrta.gov.cn/art/2021/10/20/art_113_58228.html.

[10] See Technical Requirements for Digital Virtual Humans, https://www.nrta.gov.cn/art/2024/11/28/art_3715_69714.html.



Fig. 2. Douyin profile and live streaming scenarios of Ling Yanhuan.

4.2. Media ecological perspectives on AI virtual anchors

Building on key insights from media ecology theorists such as McLuhan, Meyrowitz, and Postman, this section applies a media ecology perspective to examine how AI virtual anchors reshape the media value chain by transforming production, distribution, and consumption.

As a starting point, McLuhan (1964) highlighted that "the medium is the message." During his time, the "medium" mainly referred to forms such as television and film. People often focused solely on the content conveyed by the medium, while overlooking the social implications and structural effects inherent within the medium itself. With the advent of AI, however, the very meaning of the medium has undergone a fundamental transformation: AI itself has become a new medium form. This shift has brought about unprecedented systemic shocks, most notably reflected in the emergence of AI virtual anchors. The role once undertaken by human anchors is now performed by algorithmically generated virtual figures, leading to profound changes in the media ecology and its ethical landscape.

As Meyrowitz (1985) noted, new media not only reshape the flow of information but also create new "social situations" that redefine divisions of labor, namely, who occupies the "frontstage" and who is relegated to the "backstage." In the context of live-streaming, the virtual anchor performs visibly on stage, while human labor and technical teams behind the scenes. Audiences, in turn, enter a novel performative space co-constructed by humans and machines. In this process, the allocation of social roles and the boundaries of interaction are fundamentally reconfigured.

At the production stage, the medium shifts from "human" to "AI," replacing traditional forms of human creativity but also challenging established frameworks such as labor law and copyright law.

At the distribution stage, platforms determine the visibility of content through algorithms and moderation mechanisms, leading to a concentration of computational resources and audience traffic among top anchors. Simultaneously, the opacity of these algorithms introduces risks such as "filter bubbles." Furthermore, the absence of a settled AI ethical framework exacerbates these issues.

At the consumption stage, as Postman (1985) cautioned in Amusing Ourselves to Death, new media often allow entertainment logics to dominate serious public discourse. Although his analysis was primarily focused on television and film, the insight remains strikingly relevant in the AI era: audiences continue to consume primarily entertainment-oriented contents, and the rise of AI anchors may not altered this tendency. On the contrary, it may intensify its drawbacks and place new demands and challenges on regulation.

In sum, the development of AI virtual anchors demonstrates that new media forms themselves have become the "message." They not only restructure the media ecology of communication but also pose significant new challenges for law, ethics, and governance.

4.3. Value chain restructuring: a tripartite disruption

By substituting human spontaneity with algorithmic efficiency, transforming labor dynamics through synthetic performance, and enhancing user engagement via immersive technologies, AI virtual anchors are profoundly reshaping the media value chain and challenging the traditional "people–goods–scenarios" model at its core. This tripartite disruption not only redefines the roles of creators, platforms, and users, but also amplifies regulatory, ethical, and economic tensions within the live streaming ecosystem.

In traditional live streaming models, the value chain revolves around three core elements: people (including human hosts, operational teams, multi-channel network (MCN) agencies, and consumers), products (comprising suppliers, brands, and advertisers), and

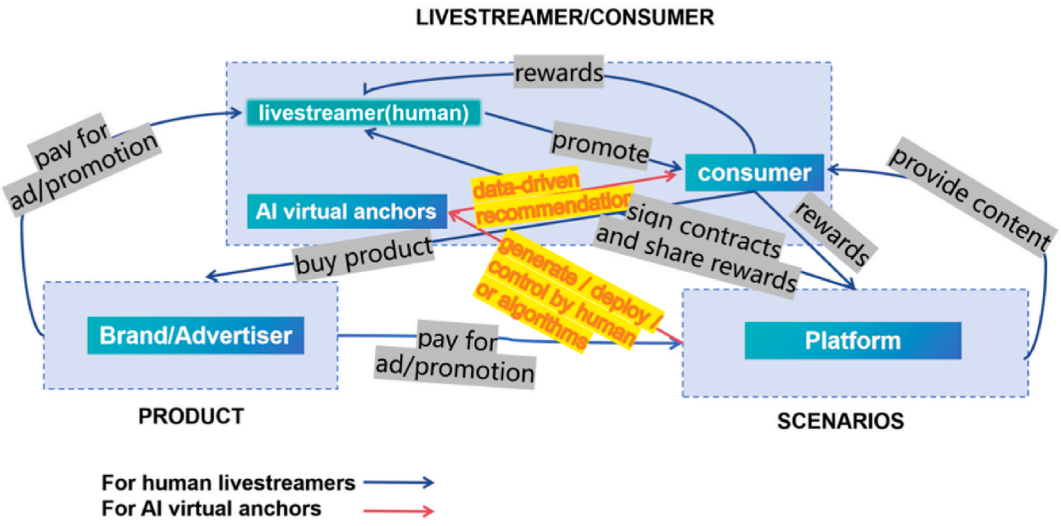


Fig. 3. The "People-Product-Scenarios" media ecosystem of live streaming.

the live streaming scenarios (represented by online platforms). In the "people–goods–scenarios" (人—货—场) model, "people" encompass both consumers and livestreamers, reflecting their central role in connecting products and platforms. Livestreamers shape emotional atmospheres that influence consumers' tipping and engagement behaviours (Lin et al., 2021; Mei et al., 2025), while consumers' real-time interactions dynamically reshape content and drive recommendation algorithms (Chen & Liu, 2024). Together, they form the emotional and behavioral core of the live streaming process, as illustrated in Fig. 3. The integration of AI virtual anchors has introduced new occupational roles, as they are not real human, they need operational teams, technical support units, and sometimes "ghost performers" (human operators who control AI avatars) for help. This transformation not only alters labour dynamics but also reconfigures the media value chain across three fundamental dimensions: content production, distribution, and consumption.

4.4. Content production: algorithmic automation and creative labour

Traditional live streaming relies on human charisma and the operational efficiency of support teams to drive content quality and commercial success (Zhang, 2021). In contrast, AI virtual anchors disrupt this model by enabling 24/7 algorithmic broadcasting at significantly lower costs (iResearch, 2024). For instance, JD.com reported that its Yanxi digital human operates at only one-tenth the cost of a human host (Li, 2024), with industry data showing a 1.93-fold increase in conversion rates and a 7.5-fold rise in daily Gross Merchandise Volume (GMV). Despite these advantages, AI anchors face persistent limitations, including delayed responses, repetitive interactions, and post-sale risks, highlighting the challenges of replicating human adaptability (Liu et al., 2025).

These developments also raise complex questions about authorship and creativity. Although AI virtual anchors exhibit expressive and generative capabilities, the originality of their output and its ownership remain legally and ethically ambiguous. China currently lacks clear regulations on AIGC, creating uncertainty around the rights of developers, users, and contributors to training datasets. Moreover, many AI live streamers rely on "ghost performers" for voice and motion control, drawing attention to the undervalued labour that supports virtual personas.

While these issues have been acknowledged in academic discussions, they are much more complicated in practise. The subsequent chapter examines representative legal cases to explore unresolved dilemmas surrounding copyright, authorship, and performance rights, offering insight into the governance challenges posed by AI-driven content creation.

4.5. Content distribution: algorithmic gatekeeping and global fragmentation

In AI-driven live streaming, platform algorithms serve as powerful gatekeepers, dictating the visibility, targeted audience, and frequency of content dissemination. AI anchors leverage profiling data, such as user age, gender, location, and behavioral patterns, to deliver personalized experiences. JD's virtual anchor Xiaomei, for example, uses real-time analytics to recommend customized products, enhancing engagement and purchase conversion.

However, this precision comes at a cost. Algorithms tend to favour high-performing content, reinforcing the dominance of top-tier streamers, human or AI, and marginalizing smaller creators (Huang & Wu, 2020; Li & Liu, 2023). This dynamic creates a self-reinforcing "Matthew Effect" that undermines content diversity and distorts competition, which challenges fairness and transparency, especially under China's Interim Measures on Generative AI (2023).

The global reach of Chinese live streaming platforms extends beyond national boundaries, raising regulatory concerns. E-commerce is promoted abroad, but it may clash with local laws. A notable example is Indonesia's 2023 ban on using social media for e-commerce, which abruptly halted TikTok's operations and affected millions of sellers and creators (Xu, 2023). The policy cited concerns over

predatory pricing and the erosion of local businesses, highlighting the geopolitical complexities of regulating cross-border live-streaming.

4.6. Content consumption: immersive engagement

AI virtual anchors are reshaping content consumption by offering immersive, interactive experiences. Leveraging technologies such as natural language processing (NLP), augmented reality (AR), and virtual reality (VR), they enable real-time dialogue and personalized engagement. In online shopping, platforms like Taobao utilize AI anchors to simulate product trials, thereby enhancing user engagement and boosting purchase confidence.

However, this technological empowerment raises significant ethical concerns. AI anchors heavily depend on user data collection and algorithmic optimization, prompting questions about transparency, consent, and data governance. Although recent regulatory frameworks have introduced principles of data minimization and informed consent, challenges persist. The opaque nature of algorithmic decision-making and limited user awareness of how their data is utilized or monetized exacerbate risks related to autonomy, trust, and digital accountability.

5. Legal and ethical risk in AI live streaming

The legal disputes surrounding AI virtual anchors can be grouped into four analytical categories. First, IPR disputes involve ownership of AIGC and the protection of human likeness, challenging the traditional concepts of authorship and originality. Second, labor rights disputes concerning ghost performers, highlight the problem of invisible labor and the limits of existing labor law, while also engaging personal privacy issues regarding the use of their biometric data. Third, issues regarding consumer protection emerge, including identity manipulation, false representation, and risks associated with data training. These issues often stem from deceptive advertising practices or algorithmically enhanced deception, both of which are fundamentally supported by data practices that involve collecting and processing personal information for model training without sufficient consent. This raises profound challenges for consumer protection, advertising law and data privacy law. Fourth, the focus on the protection of vulnerable groups highlights manipulative and addictive AI interactions that disproportionately target minors and senior citizens, thus presenting unique legal challenges under minor safeguarding regimes and raising serious ethical questions about social responsibility.

This demonstrates that AI virtual anchors do not merely present isolated legal issues, but systematically challenge and interlink established boundaries in copyright, labor, consumer protection, privacy, and child safeguarding. This convergence creates a risk landscape of unprecedented complexity and severity, which the subsequent sections will analyze in detail.

5.1. Ownership, copyright infringement and ghost performer right

The rise of AI virtual anchors has transformed live streaming e-commerce, replacing traditional content creation with algorithm-driven production (Li, 2024) and raising complex legal issues (Tang et al., 2022), especially regarding copyright ownership and liability (Xie et al., 2022). Three key challenges emerge: the legal status of AIGC, the attribution of authorship, and the rights of ghost performers.

Under Article 13 of the *China Civil Code,*[11] AI entities are devoid of legal personhood, thereby precluding them from holding copyrights or bearing liability. Responsibility for infringement, such as copyright violations or misleading advertising, is assigned to human actors, including developers, platform operators, or brands (Articles 127, 1065, 1068). In practice, three main stakeholders assert ownership over virtual anchors:

Technical Developers (Production Parties). Companies responsible for building the 3D modelling, voice synthesis, and AI behavioral systems of virtual anchors usually claim ownership over the core technical components, which are protected under Article 11 of the *Copyright Law*[12] and Article 6 of the *Patent Law*.

Platforms (e.g., Bilibili, Huya, and Kuaishou)commercialize avatars as branded content or license them to creators. For example, Bilibili operates the virtual presenter group “22/7”, while Huya manages avatars “Shuang Mawei”, utilizing them as branded entities within their respective content ecosystems.

Brand/third-party production studios co-own rights based on contractual terms for commissioned virtual anchors. Depending on contractual terms, copyright ownership may be jointly held by the brand and the production company.

However, current laws provide limited guidance on AI-authored content, resulting in ambiguity when multiple actors contribute to the development and utilization of a single avatar (Suzor, 2019). The absence of legal clarity hampers the identification of accountable parties and liability attribution (Sun, 2023).

Copyright Infringement in AI Live Streaming Ecosystems. The deployment of AI virtual anchors inevitably introduces complex, multi-tiered copyright risks across their entire operational lifecycle, from training to live performance, due to the frequent use of third-party intellectual property (IP). These risks manifest in three key dimensions:

Content Creation. AI training datasets often contain unlicensed material, such as music, artwork, or video clips, sourced without authorization. This introduces latent infringement risks at the foundational level of model development.

[11] See China Civil Code, https://www.gov.cn/xinwen/2020-06/01/content_5516649.htm.
[12] See China Copyright Law, https://www.gov.cn/guoqing/2021-10/29/content_5647633.htm.

Content Adaptation. AI systems can remix or algorithmically transform protected works into new forms, such as scripts or visual elements. These actions may violate adaptation rights under Article 10 (14) of the Copyright Law, which prohibits unauthorized derivative use.

Content Dissemination. AI-generated broadcasts may include copyrighted elements that infringe communication rights under Article 10 (12). Additionally, platforms hosting such content may face secondary liability under Article 38 of the E-Commerce Law if they fail to prevent or address infringements of intellectual property rights.

A representative case is Shinchoho Co., Ltd. v. Hangzhou Jellyfish Intelligence Technology Co., Ltd. (2024),[13] in which a virtual anchor unlawfully used a chibi-style Ultraman character. The Hangzhou Intermediate People's Court ruled this violated both adaptation and network communication rights, awarding ¥300,000 in damages.

This case underscores the ongoing legal tensions between the capabilities of AIGC and the extant copyright framework within China. While AI technologies are capable of producing novel outputs through processes of remixing and transformation, current law does not provide exemptions for such derivative creations. In the absence of prior authorization from original rights holders, any modification, utilization, or dissemination of protected content, whether undertaken by a human or an AI system, remains unlawful under prevailing copyright standards.

As AI technologies become increasingly integrated into live-streaming environments, there is a growing need to refine existing legal frameworks through more precise definitions, consistent interpretations, and enforceable mechanisms to address emerging issues more effectively, such as liability and copyright protection.

Ghost Performers' Performance Rights and Privacy Risks. AI virtual anchors generally fall into two categories. One is fully algorithm-driven, often lacking emotional nuance and interactivity, resulting in rigid and impersonal performances. The second relies on ghost performers, human operators who provide real-time voice, motion, and personality inputs to animate virtual avatars. These contributions significantly enhance realism and viewer engagement, especially in dynamic live streaming environments.

Despite their central role in creative input, ghost performers are typically excluded from authorship and performance recognition. Employers usually hold the legal rights, and contracts often require anonymity or non-disclosure, as seen in the A-SOUL project. In labour disputes, ghost performers have been informally acknowledged as "quasi-performers"; however, their legal status remains ambiguous.

This raises a key legal question: Do ghost performers qualify for performer rights under Chinese law? Articles 39 and 40 of the Copyright Law grant such rights to "persons who publicly perform works," but ghost performers operate behind digital facades, complicating eligibility.

A landmark case, Mofa (Shanghai) Information Technology Co., Ltd. v. Xu (2023), addressed this issue. Xu, the human operator behind the virtual idol "Ada," used motion capture and real-time voice generation. The Hangzhou Internet Court (2022) recognized Xu's contribution as a protected performance.

The court ruled: "The virtual digital human Ada is driven by a real person through real-time voice generation and motion capture by intelligent wearable devices. Ada's monologues, dancing, and other behaviors are not independently created by AI. The "performance" highly replicates the relevant expressions, movements, and voice of the ghost performer, Xu. It is not a new AI-generated performance but an extension of the real human performer's creative work."

This case clearly defined a legal boundary: when human creative input substantially shapes an AI-driven performance, it may be protected under existing law. It sets a judicial precedent in China for recognizing performer rights in AI-assisted content and offers valuable reference for global legislative efforts.

In addition to performance rights recognition, ghost performers face privacy risks, particularly due to the extensive data collection required for AI training, including voice, facial movement, and biometric inputs. Although China's *Personal Information Protection Law* (PIPL, 2021)[14] sets strict data standards, violations still persist. In fan-driven ecosystems, performers may be exposed through "unboxing" (kaihe), where their real identities are inadvertently or maliciously leaked. A well-known case involved the virtual YouTuber Kizuna, whose operator's personal information was leaked online, sparking backlash and influencing the platform regular operation.

These cases reveal that there is not a legal vacuum, but rather a gap in the interpretability of existing laws when applied to AI–human collaborative content. Recognizing the rights of ghost performers requires judicial elaboration on how traditional legal categories, such as authorship and performance, should be adapted to hybrid creative contexts. The challenge lies less in drafting new norms and more in interpreting existing ones in light of technological change.

5.2. Identity manipulation, false representation and data training risks

As AI avatars become increasingly realistic and commercialized, they give rise to a host of legal and ethical concerns, including identity manipulation and the blurring of boundaries between real and synthetic personas, thereby escalating the risk associated with data management. This section outlines three key risks: (1) portrait and reputation rights violations, (2) false advertising and identity theft, and (3) data training risks.

Violation of Portrait and Reputation Rights. AI virtual anchors that imitate public figures without consent violate Article 1019

[13] Shanghai Shinchoho Co., Ltd. sues Hangzhou Jellyfish Intelligence Technology Co., Ltd., Civil Judgment of Second Instance, December 30, 2024, online (The Intermediate People's Court of Hangzhou City, Zhejiang Province December 30, 2024) (China).

[14] See China's Personal Information Protection Law (PIPL), https://www.gov.cn/xinwen/2021-08/20/content_5632486.htm.

of the *Civil Code,*[15] which protects individuals against unauthorized use and distortion of their likeness. In 2024, an AI-generated deepfake of Zhang Wenhong, a renowned doctor, was used to promote supplements in a live-streaming, sparking public outrage and resulting in the permanent ban of the associated accounts. This case illustrates the commercial misuse of identity and the difficulty platforms face in moderating deceptive content.

On platforms such as Douyin, AI-generated celebrity avatars have appeared in live-streaming, creating the illusion that real public figures are endorsing products, misleading consumers and commercially exploiting public personas without consent. In response, Douyin launched major enforcement campaigns, banning over 30,000 accounts and shutting down 11 AI live streaming studios for identity abuse. These actions align with the *2023 Provisions on the Administration of Deep Synthesis*[16] and the *2024 draft Measures for Marking AI-Generated Content*, which mandate clear labeling of synthetic content to prevent public deception and strengthen accountability.

False Advertising and Identity Theft. AI-generated personas have been used to deceive consumers by imitating real individuals or fabricating fictional identities to promote products, blurring the line between authenticity and deception. A well-known case involved Ukrainian YouTuber Olga Loi Ek, whose identity was used without consent in a Douyin live streaming. The AI avatar, portrayed as a Russian girl, Natasha, mimicked Olga's appearance and promoted products to over 200,000 followers, constituting a clear violation of China's Advertising Law on misleading consumers (Articles 4 and 28).

In response, Douyin cracked down on black-market practices, banning high-profile AI accounts, such as "Russian Nana," and issuing public statements on content regulation. These incidents have raised growing concerns about the misuse of digital identity and prompted more vigorous enforcement of existing rules.

Regulatory frameworks, such as *the Opinions on Further Consolidating the Main Responsibilities of Online Platforms for Information Content (2021)* [17] and the *Interim Measures on Generative AI (2023),* require platforms and AI service providers to take full responsibility for the integrity of content, legal compliance, and the ethical use of AI.

Data Security Risks in AI Model Training. The development of AI virtual anchors relies heavily on the utilization of extensive datasets, raising serious concerns over the unauthorized use of personal data during model training. The absence of precise consent mechanisms and transparent data governance poses a significant threat to data privacy and security. High-profile incidents involving LinkedIn (BBC, 2024) and Amazon (NBC, 2023) revealed AI models trained on user data without consent, underscoring the challenges of existing protective measures.

China has faced similar controversies. In 2024, Kingsoft (WPS Office) stated in its user agreement that uploaded documents could be used for AI training after "desensitization," triggering widespread public criticism. Though the company reacted swiftly and apologized, promising that "user documents would not be used by third parties for AI training" (Sina Finance, 2024). However, several online novelists later reported that chapter outlines and plotlines generated by Doubao AI bore striking similarities to content they previously uploaded to the WPS cloud service. These allegations raised serious concerns about implicit data harvesting and the unauthorized appropriation of user-generated content.

To address such risks, China has established a comprehensive framework, anchored in *the PIPL,* and *Data Security Law*. Complementary policies, including the *Interim Measures for AI (2023)*, introduced a "dual-filing mechanism" that requires both algorithm and service registration. By late 2024, over 240 models had been registered, with 57 services approved (Ning et al., 2024).

Article 7 of the *Interim Measures for AI (2023)* mandates lawful data sourcing, protection of personal information, and explicit user consent. Article 19 requires providers to disclose training data sources, types, and labeling standards while supporting regulatory audits. *The Basic Security Requirements for Generative AI Services* further detail obligations regarding data provenance, filtering, and compliance.

Since 2024, the CAC has launched enforcement campaigns, including the "Clean and Bright Campaign" and "AI Misuse" initiatives, targeting algorithmic abuses and illegal data practices. Common violations included unauthorized scraping, use of falsified or infringing content, and lack of data inspection protocols. Unlike the ex-post regulatory model prevalent in the U.S., China adopts a regulatory approach combining ex-ante approval, ongoing supervision by cyberspace authorities, and ex-post accountability. However, challenges remain, as some enterprises sacrifice data compliance for competitive speed (Qiu, 2024).

5.3. Ethical and social responsibility challenges

The integration of AI virtual anchors into live streaming commerce presents significant ethical and societal challenges (China Central Television, 2024), particularly regarding the manipulation of vulnerable groups such as the senior citizens and minors. This section highlights two key issues: (1) the creation of fabricated content and the pursuit of malicious traffic-driven manipulation, and (2) exploitation of vulnerable groups.

Fabricated Content and Malicious Engagement. To drive traffic and monetization, some AI live streaming accounts have

[15] No organisation or individual may infringe upon the portrait right of others by means of distortion, defacement, or forgery using information technology. Without the consent of the portrait right holder, the portrait shall not be made, used, or publicly disclosed, unless otherwise provided by law.

[16] See the Provisions on the Administration of Deep Synthesis of Internet Information Services, https://www.gov.cn/zhengce/zhengceku/2022-12/12/content_5731431.htm.

[17] See Opinions on Further Consolidating the Main Responsibilities of Online Platforms for Information Content, https://www.gov.cn/zhengce/zhengceku/2021-09/16/content_5637727.htm.

fabricated sensational narratives. A prominent incident in 2022 on Bilibili involved a virtual anchor, operated by human, who falsely claimed to have escaped human trafficking, garnering over seven million views before the deception was uncovered. In response, Bilibili permanently banned the account and implemented stringent measures against the profit-driven MCN agency responsible (Pengpai, 2022). *The Regulations on the Administration of Information Content Services for Internet Live Streaming Marketing (2020 Draft)*[18] explicitly prohibit such deceptive tactics, whether human or AI generated. Platforms like Douyin, Taobao, and Kuaishou have intensified their content moderation efforts. In 2024, Kuaishou halted traffic boosts for AIGC to prioritize authenticity and real-time interaction.

Exploitation of the Senior Citizens and Minors. AI virtual anchors have also been used to exploit emotional and cognitive vulnerabilities in elderly users, posing as affectionate figures or falsely promising benefits for engagement. Simultaneously, minors have been exposed to sexualized content, violent language, and reward-based tactics to encourage tipping or purchases. Such exploitative strategies emphasize the urgent need for stricter content regulation, age verification measures, and increased accountability for platforms operating within AI-driven environments.

Despite the existence of legal frameworks such as the *Law on the Protection of* Minors (2020) and the PIPL (2021), enforcement predominantly remains reactive rather than preventive. Additionally, current content moderation technologies are inadequately equipped to identify subtle forms of abuse involving AI avatars.

As AI-generated personas become increasingly interactive and emotionally persuasive, the ethical boundary between performance and manipulation becomes increasingly blurred. Without proactive regulation and stronger platform accountability, AI-driven live streaming risks amplifying ethical dilemmas and undermining public trust in digital ecosystems.

6. Multi-tiered governance for AI virtual anchors: a government-guided regulatory framework, platform governance, and public participation

China's governance of AI virtual anchors is built on a multi-tiered framework integrating government regulation, platform responsibility, and public participation. The model embodies a government-guided, with coordinated engagement from platforms and public participation, aiming to ensure effective oversight and adaptive response to the rapidly changing landscape of AI systems.

6.1. Governmental top-level design: philosophy and regulation

At its core, China's governance reflects the dialectical unity of principle and flexibility. On one hand, laws such as the *Cybersecurity Law, Data Security Law, and PIPL* provides a stringent legal foundation. On the other hand, tools like pilot programs, dynamic content block-lists, and targeted campaigns (e.g., the Clean and Bright Campaign by CAC) facilitate responsive and scenario-specific regulation. When it comes to the AI areas, *the AI Governance Principles (2019)* established key governance objectives, including ensuring AI safety and reliability, as well as fostering sustainable economic and social growth. China seeks to balance technological advancement

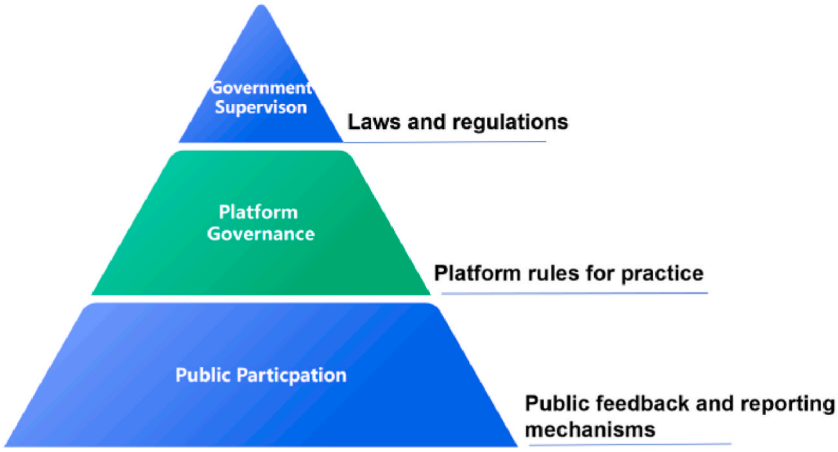


Fig. 4. Multi-tiered governance of AI virtual anchors in China. *Note:* The pyramid illustrates the multi-tiered structure: the top layer (government) establishes the regulatory framework, the intermediate layer (platforms) implements and enforces the rules, and the foundation (public) embodies the most extensive level of participation and feedback.

[18] See China The Regulations on the Administration of Information Content Services for Internet Live-Streaming Marketing (Draft for Soliciting Opinions), https://www.gov.cn/xinwen/2020-11/14/content_5561455.htm.

with security concerns (Zou and Zhang, 2025) through the guiding principles of encouraging innovation and promoting coordinated governance (Roberts et al., 2022). *Generative AI Interim Measures (2023)* propose the strategy of "addressing urgent needs first", prioritizing the resolution of compliance issues in emerging scenarios (Zhang, 2023). These mechanisms demonstrate China's effort to establish a structured and adaptive regulatory system that aims to enhance digital governance capacity while promoting inclusive access to AI technologies and minimizing systemic risks (Shen, 2024).

China's governance structure for AI live-streaming is characterized by a vertically integrated and functionally differentiated architecture that combines top-level policy design with multi-level implementation and public participation (Wu et al., 2020). The central government sets strategic objectives aligned with the national long-term plan, while regulatory agencies, such as the CAC, put them into sector-specific administrative rules. Platforms bear primary responsibility for compliance, such as algorithm filing, content moderation, and identity verification, under a "main responsibility system." Public participation, via reporting and feedback mechanisms, reinforces institutional enforcement. This model reflects a distinctive governance arrangement, combining government regulation, platform accountability, and public participation (Jian, 2024). Fig. 4 illustrates this multi-tiered governance structure in which the central government formulates the overarching legal and regulatory framework. Platforms operate at the operational level, translating national policies into actionable rules and compliance mechanisms. Meanwhile, the public participates through reporting and feedback channels, contributing to the enforcement and refinement of AI governance practices.

6.2. *Platform governance: from passive intermediaries to embedded regulators*

Gillespie (2018), in his analysis of U.S.-based platforms such as Twitter, Facebook, and YouTube, highlights how these companies initially framed themselves as "neutral intermediaries" (Gillespie, 2010) but gradually became active agents in content governance. Despite this shift, their regulatory roles are often undertaken with hesitation, largely due to the tension between governance responsibilities and commercial imperatives. In this context, platforms tend to resist regulatory obligations, particularly when such duties may compromise their market positioning or user autonomy (Nurik, 2019; Gillespie, 2018).

While Gillespie's model offers valuable insights into platform governance, its application requires contextual adaptation when examining the Chinese digital ecosystem, where platforms are more directly embedded within government-guided regulatory frameworks. Most platforms in China are similarly profit-driven and market-oriented (Sun, 2022; Wang & Zhou, 2019). However, their growing influence, evident in high-profile incidents such as Didi, was fined 8 billion RMB for unauthorized cross-border data transfers in violation of the *Cybersecurity Law, Data Security Law, and PIPL*.[19] Similarly, food delivery platforms have faced criticism for algorithmic exploitation of delivery workers. Through data-driven management and virtualized control, these systems obscure employer responsibility while tightening control over labor, leaving couriers with minimal space for resistance or autonomy (Chen, 2020).

As platforms accumulate millions of users' data and content, mobilizing vast production resources, they have become the primary targets and central actors in AI governance (Zhang, 2024). This regulatory focus has intensified since 2016, when the *Cybersecurity Law* was officially enacted. For the first time, it introduced the concept of "network operators" (网络运营者), requiring them to comply with laws and administrative regulations, respect social morality and business ethics, act in good faith, fulfill cybersecurity protection obligations, accept supervision from both the government and the public, and bear social responsibilities (Article 9), laying the legal foundation for holding platforms accountable within the digital ecosystem. Building on this foundation, the *2019 Provisions on the Governance of the Online Information Content Ecosystem* further institutionalized platforms as key governance actors. Article 2 of the Provisions promotes a regulatory model involving "government, enterprises, society, and netizens," signaling a shift away from the traditional government–market binary toward a more interactive and pluralistic governance structure. This progression reinforces the dual role of platforms as both governing entities and regulated subjects.

In the AI era, this governance trajectory has further evolved. From recommendation systems to generative models, AI technologies are now deeply embedded in platform infrastructures, reinforcing their dual role as both enforcers of regulatory mandates and subjects of external oversight. Platforms are thus tasked with implementing compliance measures such as algorithm filing, content moderation, and identity verification, while operating under the supervision of central government agencies and through public reporting channels. This structural integration gives Chinese platforms a distinctive governance function that diverges from the relatively autonomous role of their U.S.-based counterparts.

Comparative Analysis in Regulatory Approaches across Platforms. To examine how these platforms operationalise their responsibilities, this study compares five major Chinese livestreaming platforms: Douyin, Kuaishou, Bilibili, Taobao Live, and Douyu. The comparison (see Table 4) evaluates their practices across five dimensions under requirements of AI related laws and regulations, including AI anchor qualification requirements (real-name authentication; pre-registration for AI anchors; e-commerce qualification; ban AI anchors totally driven by AI), content compliance standards (ban on infringing contents; ban on content without authorization clearly), contents identifi-cation (label on AI generated content; mark the content generated by AI), measures for handling violations (account for suspension/warning/restrict account functions/forever ban; freeze revenue; cross-platform block; transfer to the judicial authorities), user rights protection mechanisms (copyright/portrait right/privacy; Full refund for the rewards from minors), and user report situation (Report Channel: Email/Hotline; Report Resolution Time Limit (within 24h), as per regulatory updates until 2023, a truth table is presented in Table 4 for comparison.

[19] CAC. (2022). 国家互联网信息办公室对滴滴全球股份有限公司依法作出网络安全审查相关行政处罚的决定 [The CAC imposes cybersecurity review-related administrative penalties on Didi Global Inc]. https://www.cac.gov.cn/2022-07/21/c_1660021534306352.htm.

Table 4
Comparison of governance practices across major platforms.

platform	AI Livestreamer Qualification Requirements				Content Compliance Requirements		Content Identification Regulations		Measures for Violations				Users Protection		User Report	
	Real-name authentication	Pre-registration for AI livestreamers	E-commerce qualification	Ban vitual livestreamers totally driven by AI	Ban on infringing contents	Ban on content without authorization	Clearly label AI-generated content (external labeling)	Mark the content generated by AI (internal marking)	Account suspension/ Warning/ Restrict account functions/ forever ban	Freeze revenue	Cross-platform block (Violate laws/ regulations/ public order and good customs.)	Transfer to the judicial authorities	Copyright/ portrait right/ Privacy	Full refund for the rewards from minors	Report channel: Email/ Hotline	Report resolution time limit (within 24h)
Douyin	✓	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Kuaishou	✓	×	×	×	✓	✓	✓	✓	✓	✓	✓	×	✓	✓	✓	✓
Douyu	✓	✓	×	×	✓	✓	✓	✓	✓	×	×	×	✓	×	✓	×
Bilibili	✓	×	×	×	✓	✓	✓	✓	✓	✓	×	×	✓	✓	✓	✓
Taobao	✓	×	✓	×	✓	✓	✓	✓	✓	✓	×	×	✓	✓	✓	×

6.2.1. Key observations

Legal Compliance with Divergence in Enforcement. While all platforms align with national policy frameworks, the stringency of enforcement varies considerably. Douyin and Kuaishou adopt stricter protocols, and Douyin escalate serious violations to authorities. Others, such as Bilibili and Douyu, maintain more lenient standards, resulting in regulatory asymmetry.

Fragmented Norms. The absence of a unified enforcement framework across platforms permits regulatory arbitrage. For example, although Douyin and Kuaishou share content blocklists or adopt similar moderation strategies, these are informal and lack institutional coordination. However, the lack of unified enforcement standards across platforms results in fragmented governance and limits the model's full efficacy. Greater transparency and inter-platform cooperation are needed to ensure consistent rule enforcement and mitigate governance fragmentation.

Limited Transparency and User Rights Protections. Although most platforms provide basic reporting channels such as hotlines or email, only a few, like Douyin and Taobao, clearly commit to timely resolution, responding within 24 hours. Mechanisms for protecting user rights, including privacy safeguards, copyright claims, and refund policies for minors, are inconsistently applied across platforms.

These variations indicate that while user participation is formally encouraged, the lack of transparency, standardization, and clear recourse options undermines the effectiveness of redress mechanisms and limits users' ability to assert their rights.

6.3. Public participation: a key component of AI governance

In the governance of AI virtual anchors in China, public participation is an essential component of the government-guided regulatory system. It has been clearly embedded in legal and institutional frameworks and plays an irreplaceable role in practice. Article 41 of the Constitution of the People's Republic of China[20] states that "citizens have the right to criticize and make suggestions regarding any state organ or functionary." Additional laws and regulations, including the *Cybersecurity Law,* the *PIPL*, and the *Interim Measures on Generative AI*, further formalize the public's right to supervise online platforms. These regulations also require platforms to provide accessible reporting channels, reflecting the institutional endorsement of public supervision.

Public reporting has had a substantive impact in numerous real-world cases. One well-known example is the "Watch Brother" corruption scandal, in which an official was investigated for corruption after netizens noticed his luxury watches.[21] In other cases, environmental violations were exposed through public complaints.[22] In digital platform governance, Didi's investigation into suspected cross-border data transfers was also linked to public attention and whistleblowing, demonstrating the capacity of public participation to identify regulatory blind spots. These cases highlight how public reporting reflects heightened public awareness and identification to emerging risks.

To support public engagement, the government and platforms have established technical and procedural mechanisms. Regulatory bodies such as the CAC provide hotlines, email portals, and mini-programs for submitting complaints. Platforms like Douyin, Bilibili, and Douyu et al., have embedded reporting systems that allow users to report violations, including false advertising, algorithmic abuse, and other conduct that violates laws and regulations. According to the CAC China Internet Illegal and Unhealthy Information Reporting Centre, more than 227 million user complaints were processed in 2024, representing a 10.4 % increase from the previous year.[23]

This reporting infrastructure is critical in the context of AI-generated virtual anchors. As discussed in section 5, the use of deepfake technology to impersonate Dr. Zhang Wenhong in live stream product promotions was exposed through user reporting, prompting platform and regulatory responses. In a landscape with massive volumes of live stream content and rapid AI-driven content creation, broad-based public participation complements official regulations. It has become a crucial force in effective AI governance.

7. Systemic challenges in China's AI virtual anchor governance

7.1. The lagging of the legal framework: unclear subject status and liability allocation

The rapid evolution of AI virtual anchor technologies has outpaced China's existing legal framework, creating significant challenges in the attribution of rights and the determination of liability. China's current laws, rooted in a binary classification of legal subjects, do not recognize AI systems as independent entities. As a result, virtual anchors are treated as tools, leaving accountability to developers, operators, or the platforms. This gap generates a recurring cycle of "innovation–regulatory lag–risk accumulation," undermining the efficacy of governance.

Debates persist over whether AI systems should be granted limited legal subjecthood (Xiong and Jia, 2024). While some scholars

[20] See the Constitution of the People's Republic of China, https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm.

[21] People's Daily Online (2013). "表哥"杨达才被判刑14年,真是一个微笑惹的祸?[Yang Dacai sentenced to 14 years—Was it really a smile that caused the trouble?]. http://opinion.people.com.cn/n/2013/0905/c1003-22818798.html.

[22] Ministry of Ecology and Environment of the People's Republic of China (2021). 生态环境部通报生态环境违法行为举报奖励工作进展情况和典型案例 [Progress and Typical Cases of Environmental Violation Reporting Reward Work Released]. https://www.mee.gov.cn/ywgz/sthjzf/zfzdyxzcf/202105/t20210525_834374.shtml.

[23] CAC Reporting Centre (2025). 2024年全国网络举报受理情况 [2024 Annual Report on Online Complaint Handling Nationwide], https://www.12377.cn/tzgg/2025/f765b050_web.html.

advocate for new classifications to prevent a "liability black hole" (Chesterman, 2020), others argue that existing frameworks suffice if human actors are held accountable (Vladeck, 2014). In practice, China lacks unified legislation on AI liability, and courts often resort to fragmented case-by-case judgments.

The issue is particularly acute in live-streaming commerce, where the allocation of responsibility among developers, operators, and platforms remains ambiguous. A 2022 case in Chongqing exemplified this trilemma: the operator of a deceptive AI avatar was penalized, while the platform that algorithmically amplified its content faced no sanction. This reflects systemic gaps in attributing legal responsibility across actors.

Further complicating regulation is the lack of targeted legal provisions for AI anchors. The *Online Anchor Conduct Code (2022)*, issued by the NRTA,[24] mentions virtual anchors but focuses primarily on the conduct of human anchors. Without specific guidance on how to regulate AI anchors, many local authorities apply human standards indiscriminately, resulting in vague and ineffective enforcement.

Three core interrelated institutional weaknesses persist. Algorithmic opacity hinders fault identification; developers may invoke exemptions by claiming that the system was not in circulation or that defects were unforeseeable; operators may evade responsibility by citing "autonomous" algorithmic behaviour, but mostly not accepted by courts; and platforms face pressures to moderate content but often lack enough capacity to enforce vast compliance comprehensively. Together, these gaps underscore the pressing need for legal clarification and institutional reform to align governance structures with the evolving practices of AI.

7.2. Institutional challenges in the protection of "ghost performers"

As AI virtual anchors become increasingly prevalent in live streaming, a critical gap has emerged in protecting "ghost performers", the behind-the-scenes individuals providing voices, motion, and persona design. Despite their essential creative input, these contributors often remain uncredited, inadequately compensated, and insufficiently protected under current legal frameworks.

Under Article 38 of China's *Copyright Law*, performers are entitled to statutory rights. However, ghost performers, who are invisible in the final output, frequently fall outside this definition. The 2022 A-SOUL controversy highlighted this gap. Although ghost performers played a vital role in shaping virtual idols, they lacked legal recognition or contractual safeguards. A 2024 Jiangsu court ruling did acknowledge ghost performance as protected creative labour, but this remains an isolated precedent without systemic reinforcement.

This legal ambiguity not only marginalizes individual rights but also creates structural inequality in the AI content value chain, where key creative roles remain invisible and unrewarded. Economically, ghost performers are routinely excluded from profit-sharing and residual income despite their contributions to commercial success. This risks deterring professional talent, undermining trust, and threatening the long-term sustainability of AI-driven creative industries.

Addressing this issue requires integrating labour protection into China's broader AI governance strategy. This includes: (1) legally recognizing ghost performers as protected creative contributors; (2) ensuring attribution and fair compensation; and (3) introducing profit-sharing mechanisms and clearer contract frameworks.

In sum, the ghost performer dilemma reflects a deeper inertia in adapting labour law to AI-era cultural production. Bridging this gap is crucial for fairness, innovation, and ethical industry development.

7.3. Fragmented governance and implementation gaps

Although China's multi-tiered AI governance model, comprising government regulation, platform responsibility, and public participation, there still exist some problems in fragmented governance and implementation gaps in practical AI virtual anchors governance.

Inconsistent regulatory enforcement across platforms. Since 2024, the CAC has initiated campaigns, such as the "Clean and Bright Campaign," to combat algorithmic abuse and AI misuse. While these campaigns have produced measurable outcomes, content takedown, administrative penalties, and warnings, enforcement varies significantly across platforms. Leading platforms, such as Douyin and Taobao, have adopted AI-powered detection tools, content audit systems, and user reporting channels. However, regional and smaller platforms often lack the technical or financial capacity to implement these tools effectively, resulting in blind spots and inconsistent application of regulatory standards.

Excessive platform power. Platforms serve dual roles as commercial entities and quasi-regulatory actors, with the authority to both define and enforce platform rules. This institutional centralization, amplified by the use of algorithmic systems, often leads to asymmetrical enforcement dynamics (Tang, 2025). Users may face account suspensions or content removed as a result of vague or poorly defined community guidelines, with limited explanation or recourse. The appeals process remains procedurally complex and lacks transparency, making it difficult for users to defend their rights. Additionally, the technical characteristics of AIGC, such as its repetitive, emotion-neutral, or templated style, make it hard to tell whether errors made by the system or truly AIGC's errors. For example, the suspension of an AI account operated by a Shanxi merchant (Tencent News, 2025) illustrates persistent gaps in procedural clarity and user protection, raising broader concerns about the accountability mechanisms in platform-based AI governance.

Furthermore, shortcomings in public reporting and redress mechanisms. As discussed earlier, while user participation is formally

[24] See China Code of Conduct for Online live streaming Anchors, https://www.nrta.gov.cn/art/2022/6/22/art_113_60757.html.

encouraged across platforms, the actual effectiveness of public reporting and feedback mechanisms remains uneven. Although such systems are embedded within platform infrastructures and endorsed by regulatory authorities, users may encounter challenges in the process of seeking redress. This lack of procedural transparency and outcome communication limits user trust and raises questions about the substantive effectiveness of participatory governance measures. It highlights the need for clearer, more standardized redress channels that can make user participation more meaningful and reactive.

Together, these governance challenges, fragmented platform compliance, inconsistent public participation, and the absence of standardized enforcement protocols reveal a gap between policy aspiration and policy implementation. Tackling these challenges requires not only technical upgrades but also structural reforms that enhance cross-platform coordination, ensure procedural transparency, and strengthen public participation in the evolving landscape of AI-mediated media governance.

8. Discussion and analysis

The emergence and governance of AI virtual anchors in China present an important opportunity to revisit platform governance theories and examine how they operate across distinct institutional contexts. Gillespie's notion of platforms as "reluctant regulators," alongside van Dijck's model of "platform societies," is primarily rooted in regulatory contexts where platforms enjoy a high degree of autonomy and often resist formal government intervention. These theories are situated mainly in governance systems characterized by limited government involvement and prevailing self-regulatory practices by platforms.

This study adopts these theoretical perspectives, using them as a comparative lens to explore how platform governance is structured differently in China's government-guided governance framework, particularly in the area of AI virtual anchor governance. By placing these theories in dialogue with empirical practices in China's live streaming e-commerce ecosystem, the paper identifies how governance responsibilities are redistributed among the government, platforms, and the public.

China's large-scale live-streaming economy, involving hundreds of millions of users, has driven the development of a hybrid model of AI governance that integrates top-level regulatory guidance with platform-level implementation mechanisms. Platforms do not operate independently, but as part of a broader regulatory ecosystem involving government guidance and societal participation.

Platforms are responsible for both regulatory compliance, including real-name registration, algorithm filings, and AI content labeling, as well as operational governance tasks such as moderating content, managing user behaviour, and responding public feedback. This institutional arrangement positions platforms in a dual role, serving both as commercial entities and regulatory enforcers. This embedded role underscores the continued integration of platforms into China's governance framework, where their responsibilities have expanded and evolved in response to emerging demands for AI governance.

China's approach to AI governance reflects a distinctive policy trajectory grounded in core priorities such as data security, national security, and public order. Within this framework, regulatory mechanisms emphasize institutional adaptability and scenario-based responsiveness. Under government guidance, the governance process mobilizes participation from multiple sectors of society. The government develops legal and policy frameworks, platforms are responsible for implementing compliance measures and operational enforcement; while the public participates through reporting mechanisms and user complaints.

Theoretically, this approach represents a hybrid governance model that illustrates how platform governance theories can be extended to accommodate various institutional configurations. The Chinese case offers a different but complementary lens to existing frameworks that emphasize industry self-regulation and platform autonomy, demonstrating how platform governance can be embedded in a government-guided, multi-tiered structure characterized by coordinated regulation and institutional integration. In this respect, it broadens the contextual applicability of existing theories, offering new insights into how governance functions may be distributed in environments where the government plays a more integrative role.

As countries confront challenges such as algorithmic harms, AI safety and reliability concerns, and disinformation risks, China model provides a unique perspective for rethinking governance at the intersection of technological innovation and institutional legitimacy. While rooted in a specific institutional and technological context, this evolving model provides a reference point for comparative discussions on how different societies organize digital governance in the era of AI.

In light of this observation, the goal of this study is not to advocate for the superiority of one governance regime over another, but to offer a comparative perspective on how China's approach fits into the landscape of AI governance, thereby contributing to ongoing international dialogue and mutual learning. In doing so, it reflects China ongoing regulatory development while aligning with global concerns about building governance frameworks that are both locally grounded and internationally relevant, especially for ethical and inclusive AI.

9. Conclusion and limitations

9.1. Conclusion

AI virtual anchors represent a paradigm shift in the media ecosystem, and rebuilding the media chain from production, distribution, and consumption, and challenging traditional legal and ethical frameworks. China's approach, anchored in scenario-based regulation, government-guided coordination, platform responsibility, and public participation, constitutes a unique model in the global landscape of AI governance.

While structural challenges persist, including legal lag, fragmented implementation, and uneven platform enforcement, China's evolving governance model demonstrates an adaptive and context-responsive strategy. It offers an indigenous framework for addressing the socio-technical complexities of intelligent media systems.

As the global community continues to explore culturally responsive and innovation-compatible approaches to AI governance, the Chinese case offers a distinct contextual example, which provides a basis for comparative reflection and institutional learning across diverse regulatory environments.

China's evolving experience enriches ongoing global discussions by highlighting alternative approaches and regulatory mechanisms shaped by a different institutional logic. On the basis of that, it also carries broader global implications. First, it exemplifies an alternative model of AI governance by embedding platforms as quasi-regulatory actors, utilizing their technological and operational strengths to support AI governance processes. Second, China demonstrates the value of balancing development and regulation through flexible governance approaches and pilot initiatives, combining innovation incentives with regulatory safeguards to enable governance to adapt more rapidly to emerging technological challenges. Third, with over 833 million Internet users (CNNIC, 2024), effective regulation cannot rely solely on government agencies. Users feedback and complaint channels contribute to identifying risks and improving accountability, providing insight into how public participation can complement government-led regulation in AI-driven media ecosystems.

Although these insights are context-specific, they can provide points of reference for policymakers in other jurisdictions, including the EU, US, and Global South, thereby enriching comparative discussions on global AI governance.

9.2. Limitation

While this study provides a comprehensive analysis of China's governance of AI virtual anchors through a multi-tiered framework that encompasses government regulation, platform responsibility, and public participation, several limitations must be acknowledged.

First, the analysis is primarily grounded in policy documents, regulatory texts, and secondary case materials. It does not include empirical data from interviews with key stakeholders, such as government regulators, platform administrators, or ghost performers. This reliance on documentary sources ensures a systematic account of the regulatory framework but inevitably narrows the scope of perspectives considered. Future research that integrates first-hand evidence would expand the analytical range and further strengthen the theoretical claims advanced here.

Second, the study could enhance its theoretical contribution by incorporating a broader comparative analysis of international AI governance. For instance, the study does not engage with other regulatory models, such as the European Union's AI Act and the approach of the United States, which could offer valuable contrasts.

Future research could pursue cross-national empirical comparisons and explore the long-term implications of algorithmic accountability mechanisms through longitudinal studies, thereby deepening our understanding of AI governance in diverse political and institutional contexts.

CRediT authorship contribution statement

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to check grammar and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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